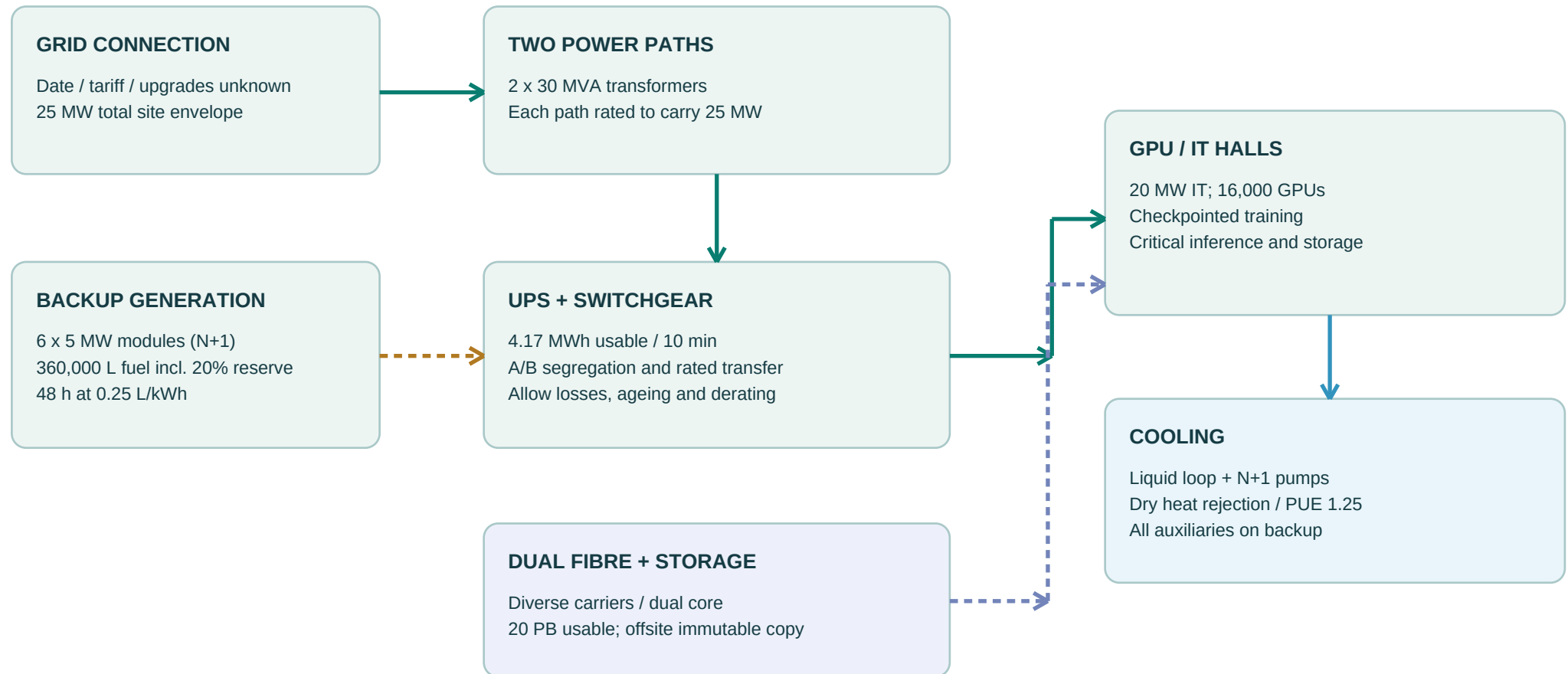


25 MW concept | Power, cooling and failure paths

Reference design, not construction-ready. Conditional phase 1 is 10 MW. All equipment ratings are assumptions.



Transformer failure: isolate path A; UPS bridges transfer; path B carries the full site. If capacity or cooling is inadequate, checkpoint and shed training before critical services.

48-hour grid loss: UPS bridges start; five generators run with one spare. Backup supports cooling and IT. If fuel or a second system fails, preserve checkpoints and shut down safely. 10 MW module: 3 x 5 MW generators, 144,000 L fuel.

Verification: protection/selectivity, fault levels, cable ratings, correlated grid failures, generator derating, fuel permits/logistics, coolant failure and heatwaves. UPS is not 48-hour storage; no renewable firm-capacity credit. Initial target 99.95% is not certified.